

4.10.16

EE25BTECH11014 - Bhoomika Lokesh

Question: The area of the region bounded by the curve $y = x + 1$ and the lines $x = 2$ and $x = 3$ is:

- 1) $\frac{7}{2} sq.units$ 2) $\frac{9}{2} sq.units$ 3) $\frac{11}{2} sq.units$ 4) $\frac{13}{2} sq.units$

Solution: $y = x + 1$ is

$$\mathbf{n}^T \mathbf{x} = c \quad (1)$$

$$\mathbf{n}^T = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (2)$$

where $\mathbf{n}^T$ is the normal vector of the line.

Let the points of intersection of the curve $y = x + 1$ and the lines $x = 2$ and $x = 3$ be **A, B**.

For **A**:

$$\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad (3)$$

The augmented matrix of the above is:

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ -1 & 1 & 1 \end{array} \right). \quad (4)$$

Then,

$$\mathbf{R}_1 \longrightarrow \mathbf{R}_2 + \mathbf{R}_1 \quad (5)$$

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 3 \end{array} \right) \quad (6)$$

$$\therefore \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}. \quad (7)$$

For **B**:

$$\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad (8)$$

The augmented matrix of the above is:

$$\left(\begin{array}{cc|c} 1 & 0 & 3 \\ -1 & 1 & 1 \end{array} \right). \quad (9)$$

Then,

$$R_2 \longrightarrow R_2 + R_1 \quad (10)$$

$$\left(\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 4 \end{array} \right) \quad (11)$$

$$\therefore \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}. \quad (12)$$

Hence the points of the quadrilateral are $(2, 0)$, $(2, 3)$, $(3, 4)$ and $(3, 0)$ (which are **A**, **B**, **C** and **D** respectively.)

Area of the quadrilateral can be calculated by using the formula,

$$Ar = \frac{1}{2} \|(\mathbf{A} - \mathbf{C}) \times (\mathbf{D} - \mathbf{B})\| \quad (13)$$

$$(\mathbf{A} - \mathbf{C}) = \begin{pmatrix} -1 \\ -4 \end{pmatrix} \quad (14)$$

$$(\mathbf{B} - \mathbf{D}) = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \quad (15)$$

$$(\mathbf{A} - \mathbf{C}) \times (\mathbf{D} - \mathbf{B}) = \begin{vmatrix} -1 & 1 \\ -4 & 3 \end{vmatrix} = -3 - 4 = -7 \quad (16)$$

$$\therefore Ar = \frac{1}{2} |-7| = \frac{7}{2}. \quad (17)$$

Therefore (option 1) $\frac{7}{2} sq. units$ is the correct answer.

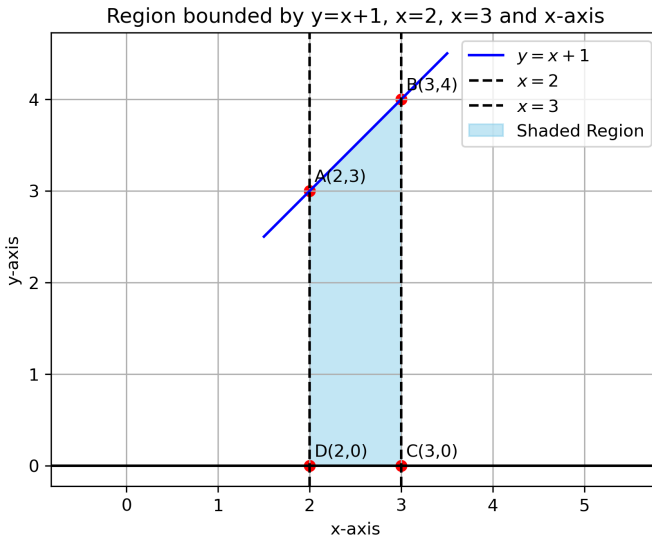


Fig. 4.1: Graph